

UGED1111 Logic

Informal Fallacies II

Recap

Fallacies of Weak Induction

- ❑ Appeal to Unqualified Authority
- ❑ Hasty Generalization
- ❑ Weak analogy
- ❑ False Cause
- ❑ Slippery slope
- ❑ Appeal to Ignorance

2. Fallacies of Relevance

Fallacies of Relevance

- ❑ Appeal to Force
- ❑ Appeal to the People
- ❑ Fallacy of Accident
- ❑ Straw Man
- ❑ Red Herring
- ❑ Argument Against the Person

2.1 Appeal to Force

The fallacy of **appeal to force** is committed whenever an arguer poses a conclusion to another person and threaten the person (implicitly or explicitly) that some harm will come to him or her if he or she does not accept the conclusion.

2.1 Appeal to Force

Example:

經理: 這裡每個人都應該要替我處理私人事務。如果你不接受，你可以來找我談談你是否應該繼續留在這間公司。

2.2 Appeal to the People

The fallacy of **appeal to the people** (訴諸群眾) occurs when the arguer tries to use the desires of being recognized/accepted by others to get the reader or listener to accept a conclusion

<https://www.youtube.com/watch?v=TzHdUP6S220>

2.2 Appeal to the People



Example:

80% of people will vote for Trump. How can you not support Trump?

2.3 Fallacy of Accident



The fallacy of accident occurs when an arguer applies a general rule to a specific case which it was not intended to cover.

2.3 Fallacy of Accident

Example:

用刀割開別人的身體，是犯罪。所以，外科醫生都犯了罪。

2.4 Straw Man

The **straw man fallacy** is committed when an arguer distorts an opponent's argument to a position which can attack more easily , It demolishes the **distorted** argument, and then claims that the opponent's argument has already been demolished.

2.4 Straw Man

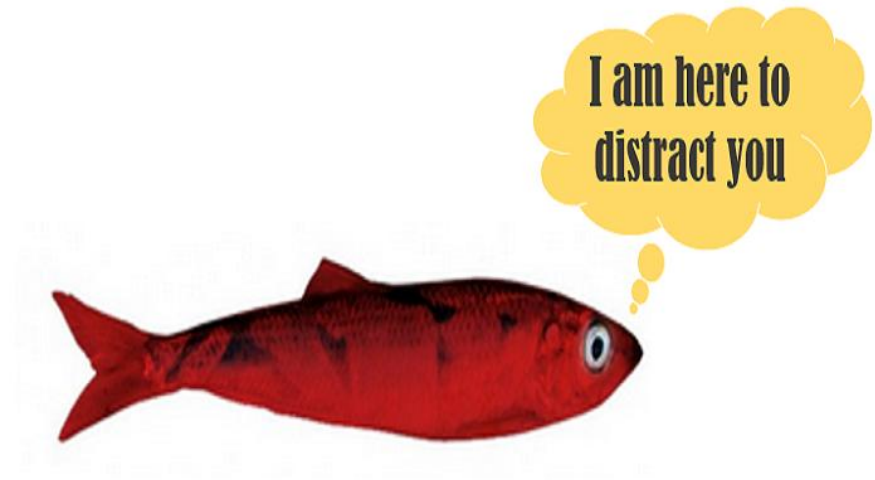
Example:

A: 我們不應該花費20億在建造一條僅200 米長的天橋，因為造價過高，而且其設計有礙景觀。

B: 為甚麼我們不應該建造天橋!沒有天橋市民會很不方便，而且那邊的車很多又沒有馬路。

2.5 Red Herring

The **red herring fallacy** is committed when the arguer diverts the attention of the reader or listener by changing the subject to a different but sometimes subtly related one.



2.5 Red Herring



A:Environmentalists are continually harping about the dangers of nuclear power and request the government to stop using nuclear power.

B:Unfortunately, electricity is dangerous no matter where it comes from. Every year hundreds of people are electrocuted by accident. Since most of these accidents are caused by carelessness, they could be avoided if people would just exercise greater caution.

2.6 Argument Against the Person

This fallacy usually involves two arguers. Arguer B responds to the argument by directing his or her attention not to the arguer A's argument but to irrelevant personal aspects of arguer A. When this occurs, arguer A is said to commit an **argument against the person**.

2.6 Argument Against the Person

a. Verbally abuse

唐唐出軌又潛建，所以他提出的所有政策都是錯的！

2.6 Argument Against the Person

b. ad hominem circumstantial

Example

John: Ms Chan argued that we should have logic lessons because we can learn how to evaluate an argument appropriately in the lesson

Peter: She said that because she teaches Logic and she needs a job! We should not believe her!

2.6 Argument Against the Person

c. “you too”

Example:

父親:吸煙有害健康，當中的有害物質增加你患癌機會，所以你最好戒煙。

兒子:笑話，吸煙有害?你自己也天天吸煙!

Discussion 1



我們應該禁止中學生談戀愛。因為談戀愛就會發生性行為，發生性行為就會懷孕，中學生又沒有經濟能力，那時候誰幫他們養孩子?!

Discussion 2



海豚和人類一樣是能分泌乳汁養育幼兒，而且都一樣是高智商生物，又同樣靠肺部呼吸，所以人類既然可以單靠食素生存，海豚也應該可以。

Discussion 3

A: 自從福島核電廠事故後，我們有理由相信福島出產之水果可能遭受污染及帶有輻射，進食福島水果可能會影響健康，所以食環署應進行輻射審查。

B: 其實香港對各國進口水果一向都有進行抽查，檢驗其是否含過量農藥，如果農藥過量定必被整批抽起，嚴禁在市面發售，日本水果作為進口水果必定有進行農藥抽查。

Discussion 4



小明 :開電單車一定戴頭盔，這樣出意外時脆弱的頭部才能受到保護。

大雄:廢話!頭盔哪有用?我上次明明就看見你沒戴頭盔就開電單車出去!

Discussion 5



現時大部份人都支持安樂死合法化，既然大家都支持，你也很應該支持。

Discussion 6



I drank 8 shot of rum and Cokes on Monday, and I woke up with a headache. On Wednesday, I drank 8 shot of whisky and cokes, and I woke up with a headache. On Friday, I drank 8 shot of gin and Coke, and I woke up with a headache. On Sunday, I drank 8 shot of Vodka and cokes and I woke up with a headache. Obviously, I should quit coke to prevent headache.

Discussion 7

John 和May 約會時發現May 因為約會沒有在高級餐廳而生氣。John 和June 約會後，June 因為John 沒有開勞斯萊斯而不再跟他約會。John 和Idy約會後，Idy 因為他沒有黑卡而斷絕來往。John 由此判斷所有女人都是貪慕虛榮。

Discussion 8



We should always be honest. Therefore, we should tell a murderer where John is hiding himself even though John will be killed.

Discussion 9



Peter : 根據多間大學的研究報告，女性在職場中仍然受到歧視，同職位的女性一般比男性工資少20%，相關的報告可以在大學網頁查看到。

John: 女性研究員所做的研究，這怎會可信？

Discussion 10



這個科學家拋棄自己的父母，又虐待自己的妻子，
這種人提出的理論怎麼可能是正確的？

Discussion 11

如果你唔留心上堂，我就會好傷心。

我好傷心，就再冇心情食飯，咁餐廳既生意就會大受大擊。

餐廳倒閉，會令到無數人失業，佢地失業之後可能會上街搶掠，社會治安就會崩潰。

所以，如果你唔留心上堂，香港治安就玩完架喇。

Fallacies



- ❑ There could be more than one fallacy committed in the argument!
- ❑ There are some fallacies which are not taught in class.
- ❑ The important point : Is the conclusion well-supported by reasons?

Case 1

小波和財團關係密切，又曾多次被揭發收取財團巨額報酬後推出有利財團之政策，又經常不跟法律做事。因此，他並不是個稱職的公務員。

***Not committed Argument Against the Person
It is relevant.***

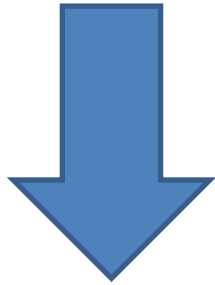
Case 2



A: You are so fat. Your BMI is over 35. Do you know you are as fat as a pig? You should do some exercise for your own good.

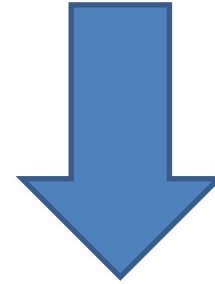
Personal Attack and Argument Against the Person

Child : I hate you! You are so ugly, stupid and fat!



Personal Attack
(人身攻撃)

Child : You are so ugly, stupid and fat, so your argument is not good !



Argument Against the Person
(人身攻撃謬誤)

Fallacy Man

Don't be a Fallacy Man!

THE ADVENTURES OF:

FALLACY
MAN!

MASTER OF PHILOSOPHY,
LORD OF DEBATE,
SULTAN OF REASON!

♪ DUN DUN DUN
DUN DUN DUNN
DUNNNNNN! ♪



Principle of charity

Principle of charity (寛容原則):

The main idea is that “in interpreting someone, we’re aiming to make sense of her; but we don’t count as making sense of someone if we represent her as being inexplicably foolish.” (Morris, 2007, p. 201)

When we encounter an argument...

We should ...

- ◆ Analyze the argument carefully
- ◆ Consider any possible hidden premises
- ◆ Pay attention to the reasoning steps
- ◆ Ask for a clarification if you find anything misleading or unclear.

Fallacies

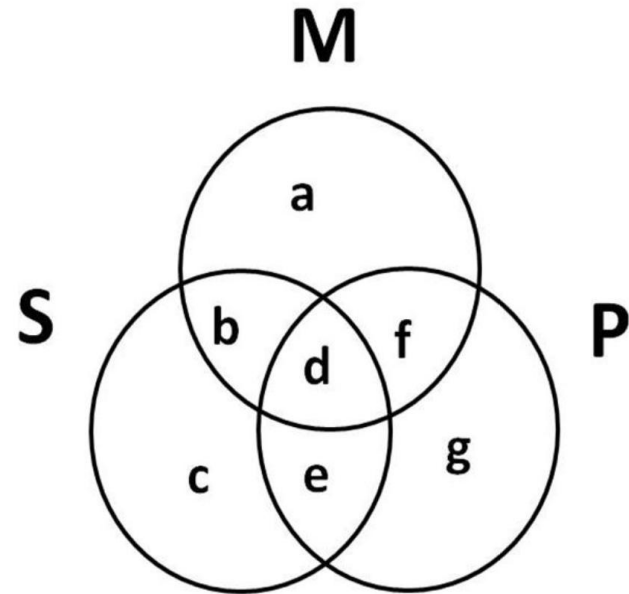
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7. Appeal to Force
8. Appeal to the People
9. Fallacy of Accident
10. Straw Man
11. Red Herring
12. Argument Against the Person
13. Complex Question
14. Begging the Question
15. False Dichotomy
16. Equivocation (not included in Exam)

Reference

- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stanford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.
- Moore, B., & Parker, R. (2012). *Critical Thinking (10th ed.)*. New York: McGraw-Hill.
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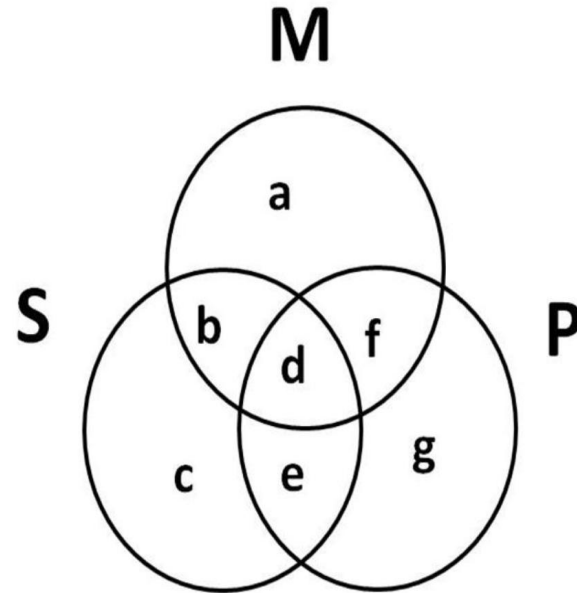
Exercise: Categorical Syllogism

1. All M are P.
 2. No S are M
-
3. All S are P.



Exercise: Categorical Syllogism

1. Some M are P.
 2. No M are S
-
3. Some P are S.



Exercise

1. $(A \vee B) \supset B$	// $A \supset (A \supset B)$
2. A	ACP
3. $A \vee B$	2, Add
4. B	1, 3, MP
5. $B \vee \sim A$	4, Add
6. $\sim A \vee B$	5, Com
7. $A \supset B$	6, Impl
8. $A \supset (A \supset B)$	2-7 CP